# Dongsung Huh

Physics of Deep Learning
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## Research Interests

Mathematical foundations of deep learning; learning dynamics and parameter sensitivity; discovery of algebraic structure and reusable computational rules; systematic generalization; theoretical neuroscience.

## Current Academic Affiliations

2026–present **Affiliate**, *The NSF AI Institute for Artificial Intelligence and Fundamental Interactions (IAIFI)*

2026–present **Visiting Scientist**, *Centaur AI Institute*

## Experience

2025–2026 **Senior Research Scientist**, *IBM Research, Mathematics of Computing*
Research on parameter sensitivity, algebraic structure discovery, systematic generalization, and uncertainty quantification.

2018–2025 **Research Staff Member**, *MIT-IBM Watson AI Lab*, Cambridge, MA
Research on learning dynamics, curvature-corrected optimization, and causal and invariant representations.

2017–2018 **Research Scientist**, *Qualcomm Institute*, San Diego, CA
Developed gradient-based training methods for continuous-time spiking neural networks.

2015–2017 **Postdoctoral Fellow**, *Salk Institute for Biological Studies*, La Jolla, CA
Advisor: Terrence Sejnowski. Studied movement power laws and conservation laws using optimal control theory.

2012–2015 **Postdoctoral Fellow**, *Gatsby Computational Neuroscience Unit, UCL*, London, UK
Advisors: Peter Dayan and Peter Latham. Studied dynamics, training, and control of spiking neural networks.

2005–2009 **Graduate Researcher**, *Optimal Motor Control Lab, UCSD*, La Jolla, CA
Advisor: Emanuel Todorov. Studied optimal control under multiplicative noise and recurrent networks for motor control.

2006–2007 **Visiting Scientist**, *Center for Brain Science, Harvard University*, Cambridge, MA
Advisors: Surya Ganguli and Haim Sompolinsky. Studied memory capacity and dynamics in dynamical systems.

## Education

2012 **Ph.D. in Physics**, *University of California, San Diego*, San Diego, CA
Thesis: *Rethinking about Optimal Control Theory of Human Movements*

2010 **M.S. in Neuroscience**, *University of California, San Diego*, San Diego, CA

2004 **B.S. in Physics**, *Massachusetts Institute of Technology*, Cambridge, MA

2004 **B.S. in Brain and Cognitive Sciences**, *Massachusetts Institute of Technology*, Cambridge, MA

## Selected Publications and Manuscripts

2026 **D. Huh**, L. Horesh, H. Jeong. Associativity as Occam's Razor: Differentiable Measure of Algebraic Complexity. *Submitted to ICLR 2027*.

2026 **D. Huh**, H. Jeong. Exact Group Recovery from Subquadratic Observations. *Submitted to ICLR 2027*.

ICLR 2025 **D. Huh**. Discovering Group Structures via Unitary Representation Learning.

2026 R. Nartallo-Kaluarachchi, R. Manson-Sawko, S. Ubaru, **D. Huh**, M. J. Zimoń, L. Horesh, Y. Bengio. Interpretable attribution of decision-relevant uncertainty in AI. *Nature Machine Intelligence, accepted*.

NeurIPS 2022 **D. Huh**, A. Baidya. The Missing Invariance Principle Found: The Reciprocal Twin of Invariant Risk Minimization.

ICML 2020 **D. Huh**. Curvature-Corrected Learning Dynamics in Deep Neural Networks.

## Selected Funded Research Projects

Co-PI **Safe AI Approaches and Program Synthesis**. MIT-IBM project with Armando Solar-Lezama (MIT). Project funding: $700,000.

Co-PI **Discovering Algebraic Structures towards Automatic Symmetry-aware AI Models**. IBM-UAlbany Center for Emerging AI Systems project with Halyun Jeong (SUNY Albany). Project funding: $100,000.

## Research Mentoring

IBM Research Mentored five doctoral researchers during summer internships, helping define research questions, meeting regularly, and guiding analyses.
Interns mentored: Joel Dapello (MIT), Avinash Baidya (UC Davis), Ramon Nartallo-Kaluarachchi (Oxford), Fan Tian (Tufts), and Paulina Hoyos (UT Austin).

## Selected Talks

Aug. 2026 **Invited Speaker**, *5th Annual Neuro-Symbolic AI Summer School*, Centaur AI Institute, Virtual
Beyond Smooth Interpolation: Differentiable Discovery of Algebraic Structures

July 2026 **Invited Seminar**, *Google DeepMind*, Universal AI Group, London, UK
Beyond Smooth Interpolation: Differentiable Discovery of Algebraic Structures

June 2026 **Invited Seminar**, *MIT*, Physics of Living Systems and Non-Equilibrium Systems, Cambridge, MA
Beyond Smooth Interpolation: Differentiable Discovery of Algebraic Structures

May 2026 **Invited Seminar**, *Flatiron Institute*, Center for Computational Mathematics, New York, NY
Differentiable Algorithmic Compression: Provably Exact Discovery of Group Structures

Feb 2026 **Redwood Seminar**, *UC Berkeley*, Redwood Center for Theoretical Neuroscience, Berkeley, CA
Beyond Interpolation: Automated Discovery of Symmetry Groups via Tensor Factorization

Dec. 2018 **Invited Speaker**, *Montreal Artificial Intelligence & Neuroscience (MAIN)*, University of Montreal, Montreal, Canada
Exploring Space of Spike-based computations

## Selected Honors & Awards

2012–2015 Gatsby Charitable Foundation Training Fellowship

2004–2006 National Science Foundation (NSF) Graduate Research Fellowship

2000–2004 T.J. Park Undergraduate Fellowship

1999 **Gold Medal**, 30th International Physics Olympiad (IPhO), Italy

## Publications

### Journal and Conference Papers

2026 R. Nartallo-Kaluarachchi, R. Manson-Sawko, S. Ubaru, **D. Huh**, M. J. Zimoń, L. Horesh, Y. Bengio. Interpretable attribution of decision-relevant uncertainty in AI. *Accepted at Nature Machine Intelligence*.

2026 J.L. Cook, D.S. Fraser, L.J. Hickman, R. Brewer, **D. Huh**. The relationship between speed and curvature differs in autistic and non-autistic tracing movements. *Scientific Reports*, 16 (1), 37067.

2025 **D. Huh**. Discovering Group Structures via Unitary Representation Learning. *ICLR 2025*, The Thirteenth International Conference on Learning Representations.

2023 F. Petersen, T. Sutter, C. Borgelt, **D. Huh**, H. Kuehne, Y. Sun, O. Deussen. ISAAC Newton: Input-Based Approximate Curvature for Newton's Method. *ICLR 2023*, International Conference on Learning Representations.

2022 **D. Huh**, A. Baidya. The Missing Invariance Principle Found: The Reciprocal Twin of Invariant Risk Minimization. *NeurIPS 2022*, Advances in Neural Information Processing Systems, 35, 23023-23035.

2021 B.A. Schuster, D.S. Fraser, J.J.F. van den Bosch, S. Sowden, A.S. Gordon, **D. Huh**, J.L. Cook. Kinematics and Observer-Animator Kinematic Similarity Predict Mental State Attribution From Heider–Simmel Style Animations. *Scientific Reports*, 11 (1), 18266.

2020 **D. Huh**. Curvature-Corrected Learning Dynamics in Deep Neural Networks. *ICML 2020*, International Conference on Machine Learning, 4552-4560.

2018 **D. Huh**, T. J. Sejnowski. Gradient Descent for Spiking Neural Networks. *NeurIPS 2018*, Advances in Neural Information Processing Systems, 31.

2016 **D. Huh**, T. J. Sejnowski. Conservation Law for Self-Paced Movements. *PNAS*, Proceedings of the National Academy of Sciences, 113(31), 8831-8836.

2015 **D. Huh**, T. J. Sejnowski. Spectrum of Power Laws in Curved Hand Movements. *PNAS*, Proceedings of the National Academy of Sciences, 112(29), E3950-E3958.

2011 **D. Huh**, T. J. Sejnowski. Exact Analytic Solutions for Optimal Control Problems Under Multiplicative Noise. *IFAC Proceedings*, 18 (1), 5908-5914.

2010 **D. Huh**, E. Todorov, T. J. Sejnowski. Infinite-Horizon Optimal Control for Goal Directed Movements. *NCM*, Neural Control of Movements.

2009 **D. Huh**, E. Todorov. Real-Time Motor Control Using Recurrent Neural Networks. *ADPRL 2009*, IEEE International Symposium on Adaptive Dynamic Programming and Reinforcement Learning.

2008 S. Ganguli, **D. Huh**, H. Sompolinsky. Memory Traces in Dynamical Systems. *PNAS*, Proceedings of the National Academy of Sciences, 105(48), 18970-18975.

### Workshop Papers

2025 R. Dineth Nartallo-Kaluarachchi, R. Manson-Sawko, S. Ubaru, **D. Huh**, M. J. Zimon, L. Horesh. Quantifying Policy Uncertainty in Generative Flow Networks with Uncertain Rewards. *NeurIPS 2025 Workshop (ML×OR)*, Mathematical Foundations and Operational Integration of Machine Learning for Uncertainty-Aware Decision-Making.

2025 **D. Huh**, H. Jeong. Theoretical Analysis of HyperCube Objective for Group Representation Learning. *NeurIPS 2025 Workshop*, Workshop on Symmetry and Geometry in Neural Representations.

### Preprints and Manuscripts

2026 **D. Huh**, L. Horesh, H. Jeong. Associativity as Occam's Razor: Differentiable Measure of Algebraic Complexity. *Submitted to ICLR 2027*.

2026 **D. Huh**, H. Jeong. Exact Group Recovery from Subquadratic Observations. *Submitted to ICLR 2027*.

2026 P. Hoyos, S. Ubaru, **D. Huh**, L. Horesh, V. Kalantzis, K. Clarkson, M. Kilmer, H. Avron. Exact Symmetry as Algebra: Group-Structured Tensors that Enforce Physical Selection Rules. *In submission to PNAS*, arXiv:2605.20440.

2025 **D. Huh**, L. Horesh, H. Jeong. A Differentiable Measure of Algebraic Complexity: Provably Exact Discovery of Group Structures. arXiv:2511.23152.

2023 J.L. Cook, D.S. Fraser, L.J. Hickman, R. Brewer, **D. Huh**. Autistic Kinematics Diverge From the Power Laws That Typically Govern Movement. *BioRxiv*.

2017 **D. Huh**. Generalized Phase Representation of Integrate-and-Fire Models. *arXiv preprint*, arXiv:1710.10428.

2015 **D. Huh**. The Vector Space of Convex Curves: How to Mix Shapes. *arXiv preprint*, arXiv:1506.07515.

## Patents

Pending **Discovering Discrete Group Symmetries from Data**. Patent pending. Reference: `P202502841US01`.

Pending **Incremental Updating of Truncated Higher-Order Singular Value Decompositions**. Patent pending. Reference: `P202500687US01`.